

PCaPAC 2018

12th International Workshop on
Emerging Technologies and Scientific Facilities Controls

October 16-19, 2018 Hsinchu, Taiwan



Maintenance and Optimization of Insertion Devices at NSLS-II using Motion Controls

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Outline

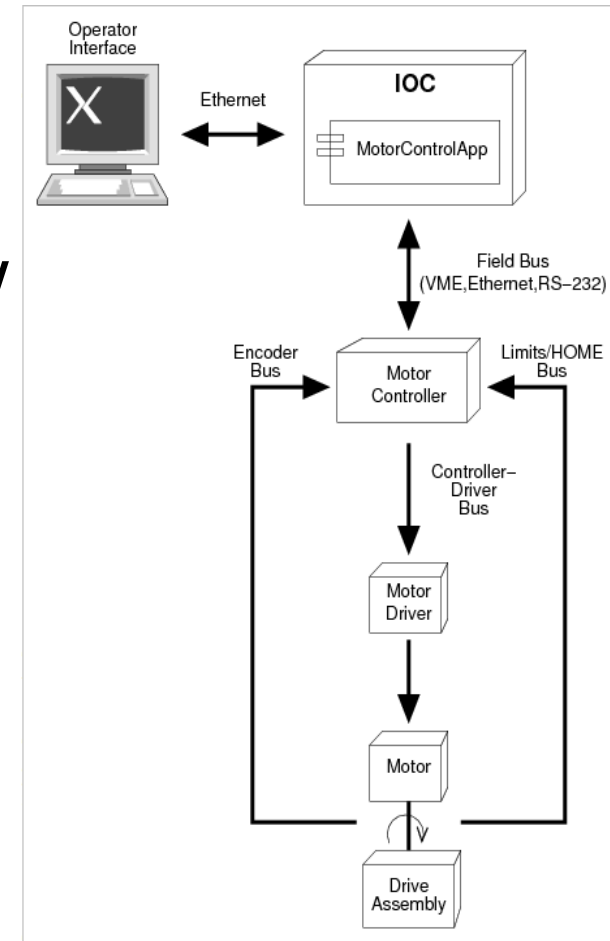
1. Preface on motion controls at NSLS-II
2. Accelerator motion controls
 1. Insertion devices (3PW, IVU, EPU, etc.)
 2. ID Issues
 3. ID Improvements
3. Q&A

Motion control hardware

- 64-bit Linux computers
- Delta Tau Programmable Multi Axis Controller (PMAC-2) and power supply modules
- Stepper/servo motors (and brakes where needed)
- Optical glass linear encoders
- Embedded rotary positional encoders (for some IDs)
- Ethernet, motor, encoder cables, etc.

Motion control software

- 3 Levels of software: controller configuration, IOC, and operator interface via EPICS communication
- Delta Tau Turbo PMAC GeoBrick LV is the standard motor controller at NSLS-II
- IOC has EPICS applications to channel access interface with motor controller
- User interfaces are OPI screens on CSS

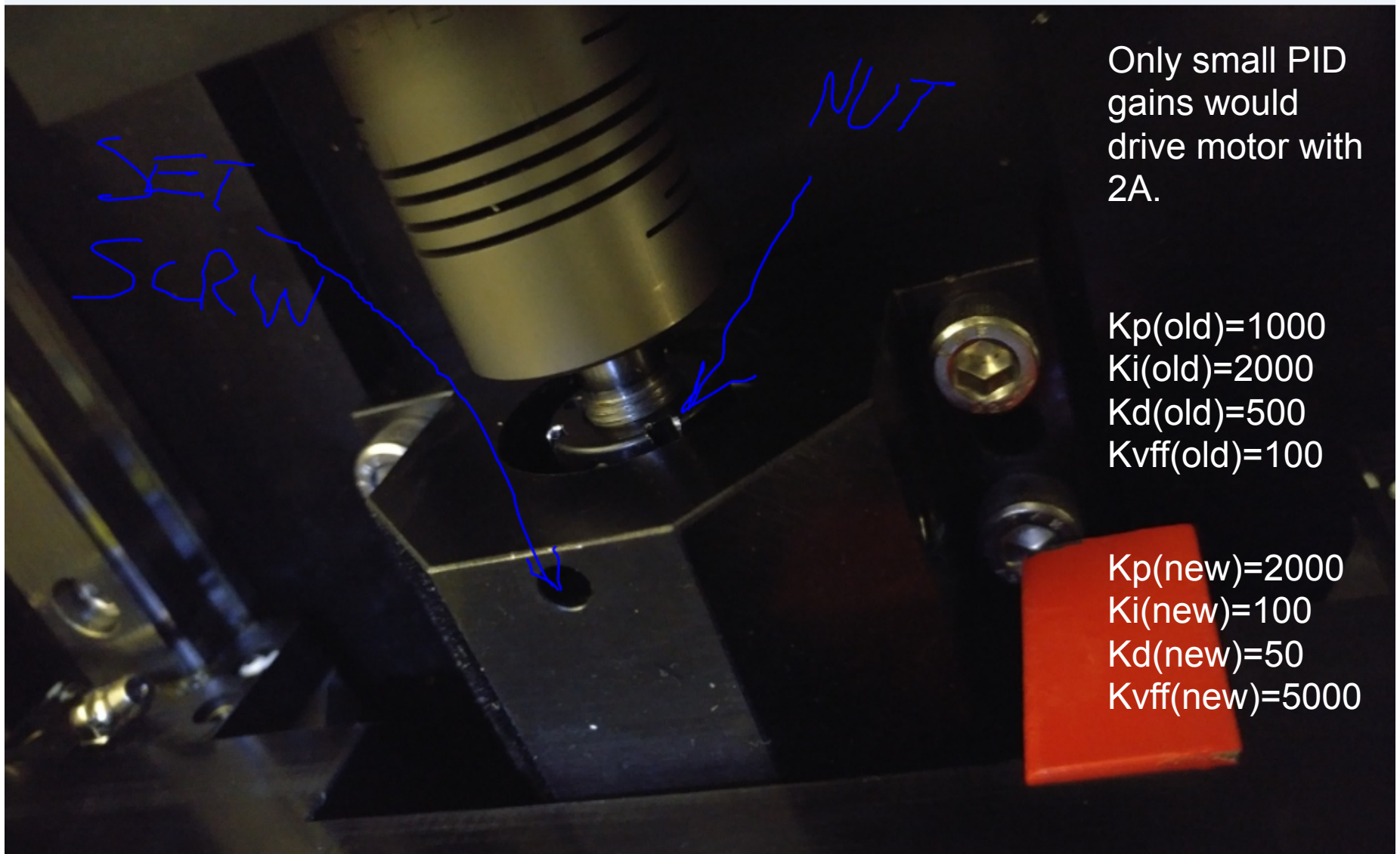


Insertion device issues

- 3PW motor moves too slow
- Undulator controls geometric issues
- EPU phase jumping
- Mechanical gear box/assemblies
- IOC server too slow

Proposed Solutions: cross couple gantry, retune PID/ adaptive tuning, ESA, migrate ID IOCs to new physical machine servers

Loose 3PW Ballscrew



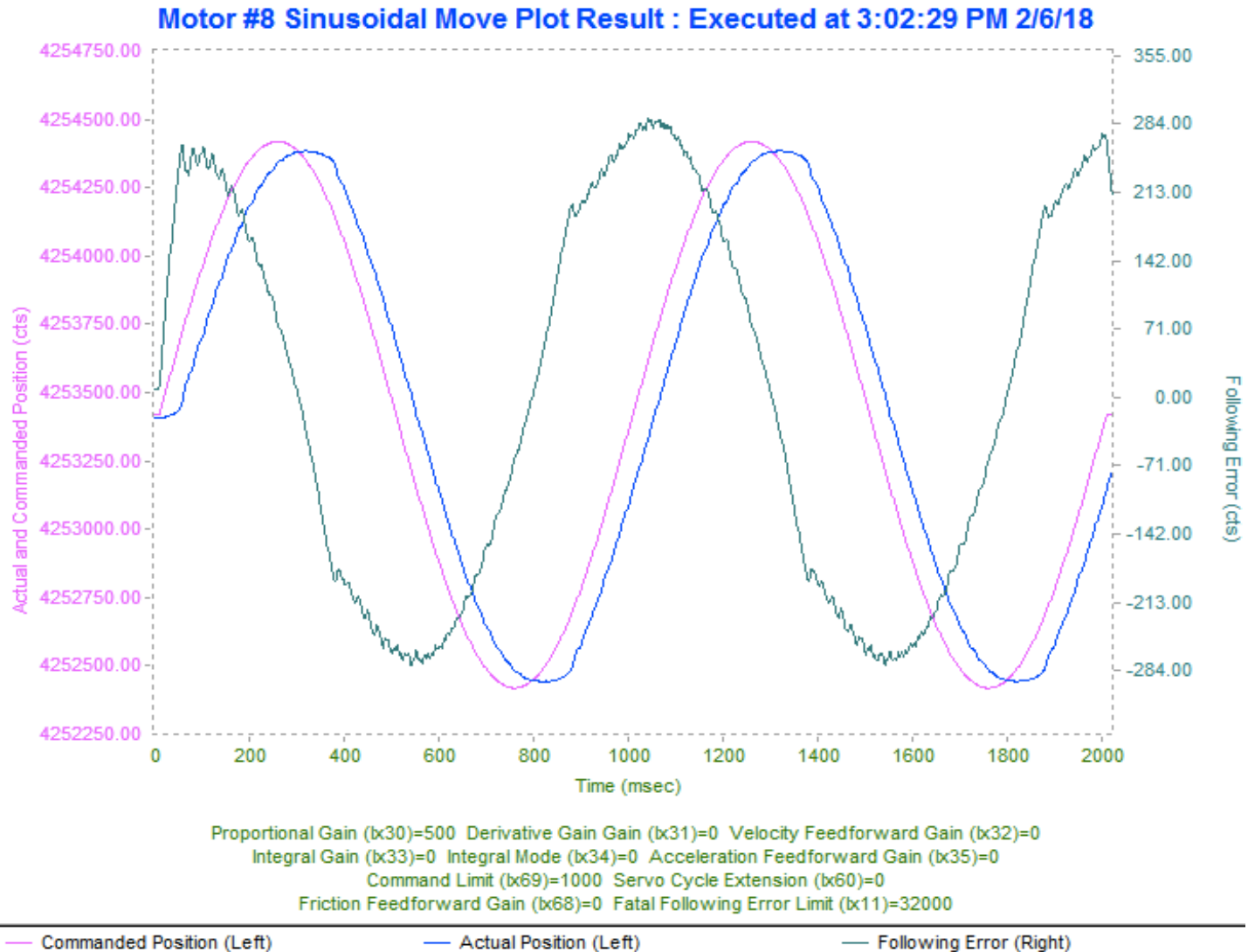
Only small PID gains would drive motor with 2A.

$K_p(\text{old})=1000$
 $K_i(\text{old})=2000$
 $K_d(\text{old})=500$
 $K_v\text{ff}(\text{old})=100$

$K_p(\text{new})=2000$
 $K_i(\text{new})=100$
 $K_d(\text{new})=50$
 $K_v\text{ff}(\text{new})=5000$



3PW Motion plot with loose ballscrew



3PW CSS OPI Screen

3PW Status

- ☒ In
- ☐ Out
- ☐ In Motion
- ☒ Homing In Process
- ☐ E-Stop kill switch
- ☐ Positive Limit
- ☐ Negative Limit

NO_ALARM

Movement

MOVE IN

MOVE OUT

Insert Limit 0.00000 mm

Retract Limit -200.00000 mm

setpoint 0.00 mm

position 0.00 mm

Stop

Pause

Move

Go

Beampipe contact sensors

DSU

USU

DSL

USL

Engineering Mode

Was running at 0.2 mm/sec (15-20mins for full motion), now running at 2mm/sec

Neomax IVU CSS OPI screen

Upper Beam

- Upper Kill #1
- Upper Limit #1
- Lower Limit #1
- Lower Kill #1

Gap

- Gap Close Limit
- Gap Close Kill
- In
- Out

Lower Beam

- Upper Kill #2
- Upper Limit #2
- Lower Limit #2
- Lower Kill #2

Evacuation

- Interlock Status

Home

Move

Alarm status

Servo ready

Servo on

Alarm reset

Units are mm

- Emergency Stop
- Evacuation Signa

Home

Move

Alarm status

Servo ready

Servo on

Alarm reset

Home

Move

Alarm status

Servo ready

Servo on

Alarm reset

Home

Move

Alarm status

Servo ready

Servo on

Alarm reset

Stop All Axes

Beam

Move Mode

- Individual Move On
- Synchronous Move On

Gap

- Set 25.0000 mm
- Actual 24.9995 mm
- Start
- Stop
- Limits 6.10 25.10

Correction Function

- Enabled
- Enable
- Disable

Vertical Offset

- Set 0.0000 mm
- Average -0.0001 mm
- Start
- Stop
- Limits -9.95 9.95

Engineering Mode

Temperature

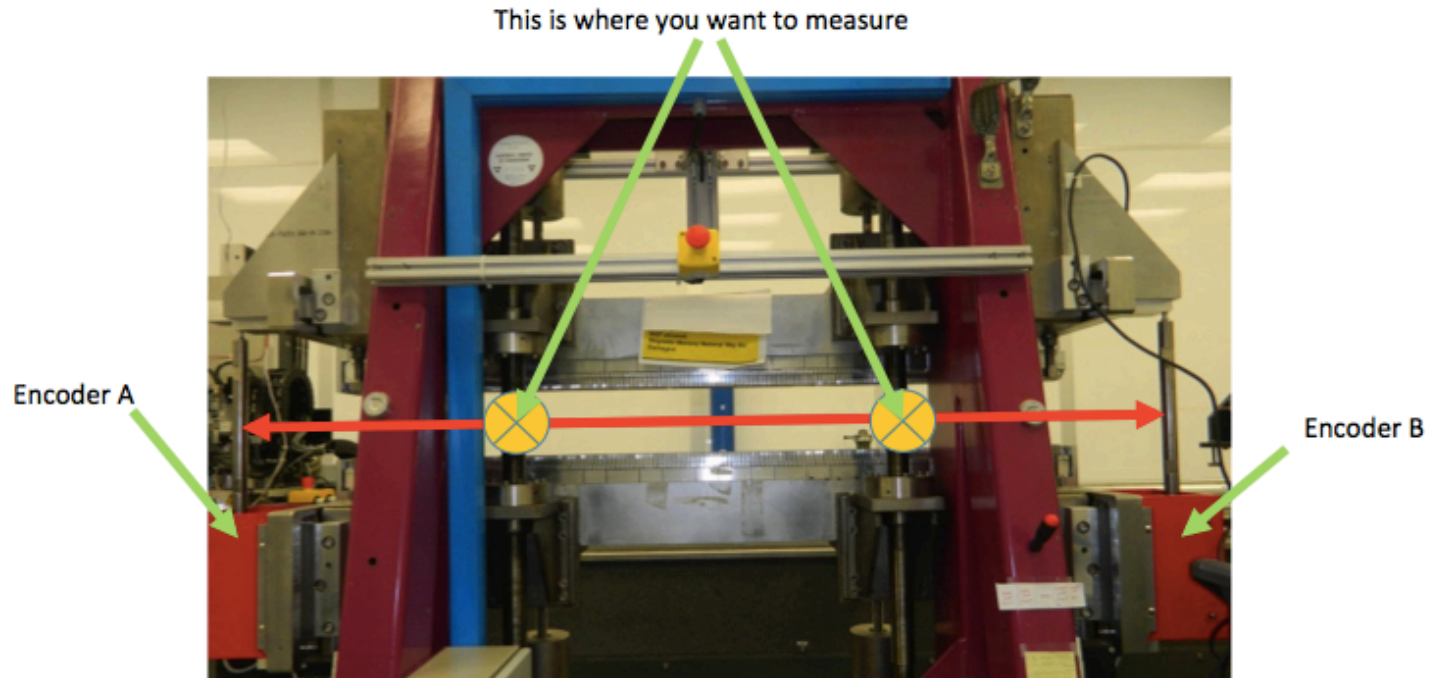
Error message

- Motor Controller: [020] Fatal following error on axis1
- PLC:

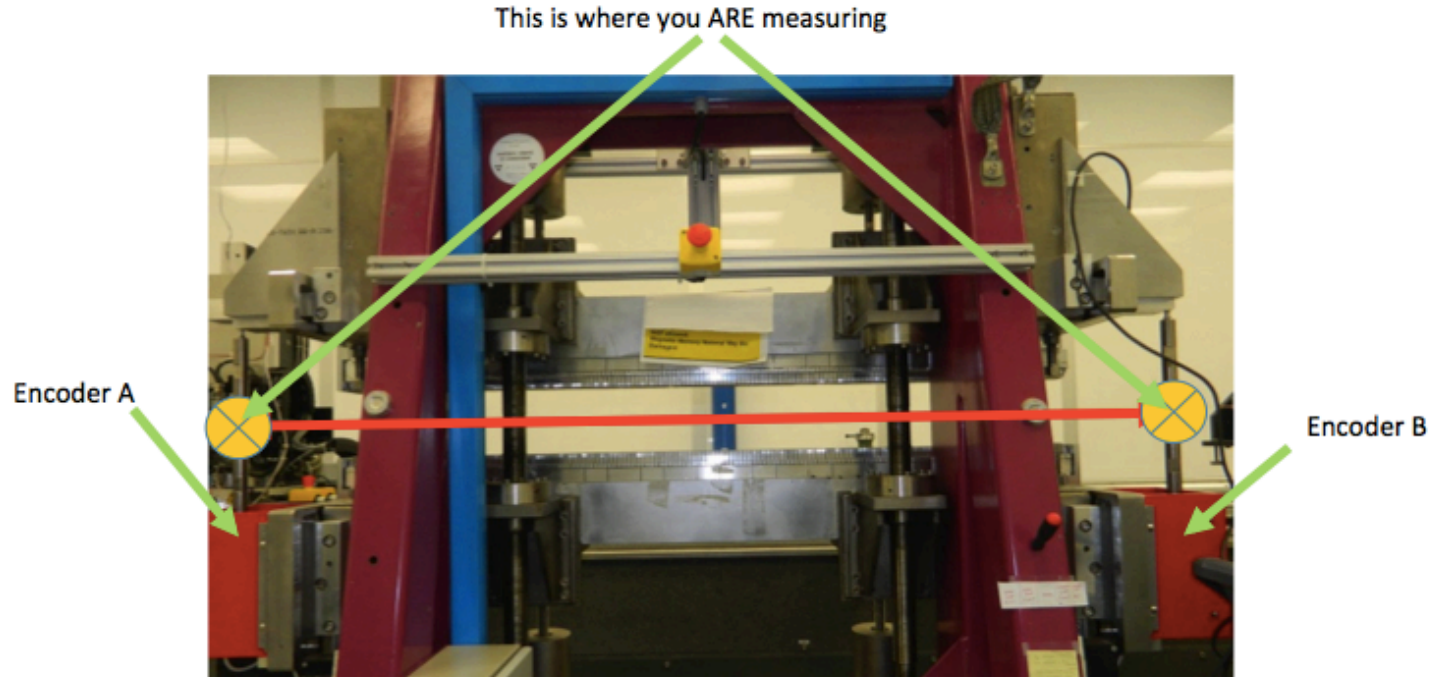
MPS

MPS Lookup

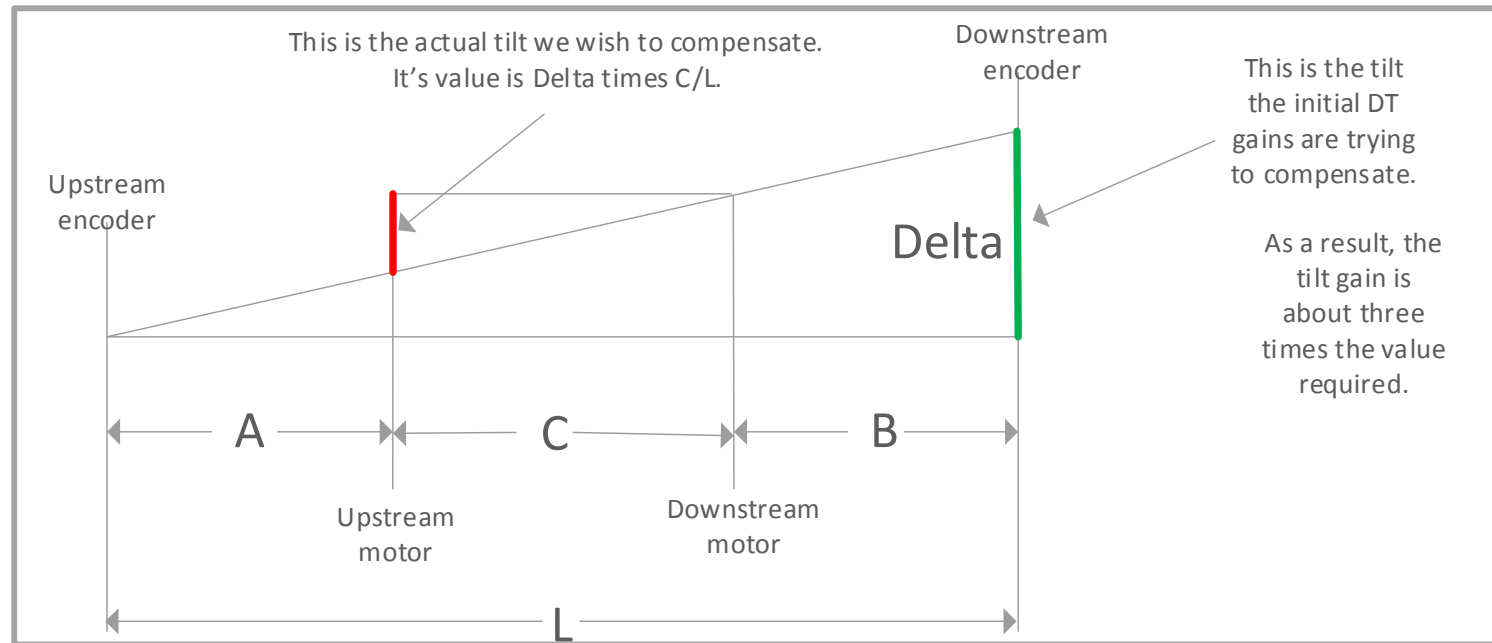
Undulator Cross-Coupled Gantry



Undulator Cross-Coupled Gantry



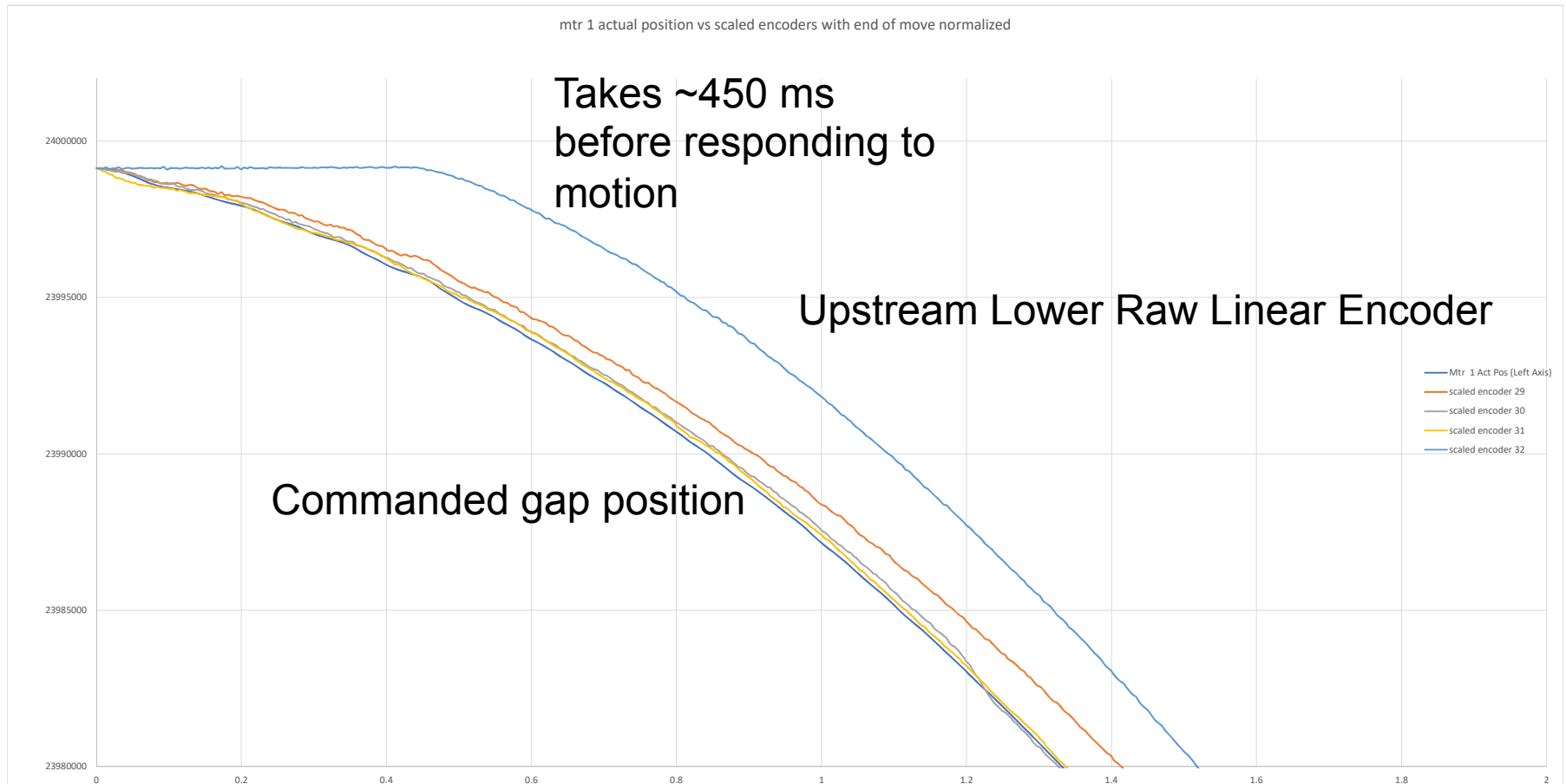
Geometry Cantilever Correction

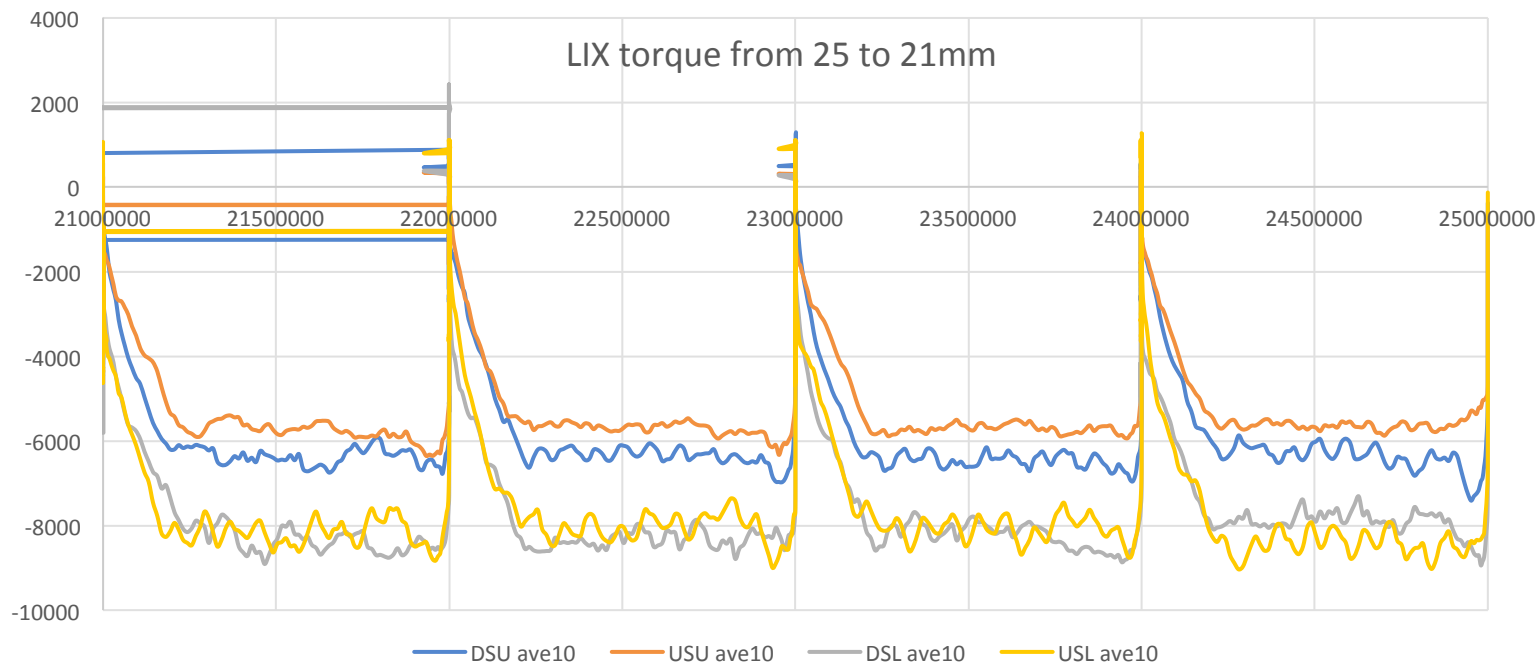


- Give true ballscrew locations
- Remove positional error from cross couple gantry algorithm
- Allow higher proportional gain to reduce following error
- Eliminate need to install encoders directly on ballscrews

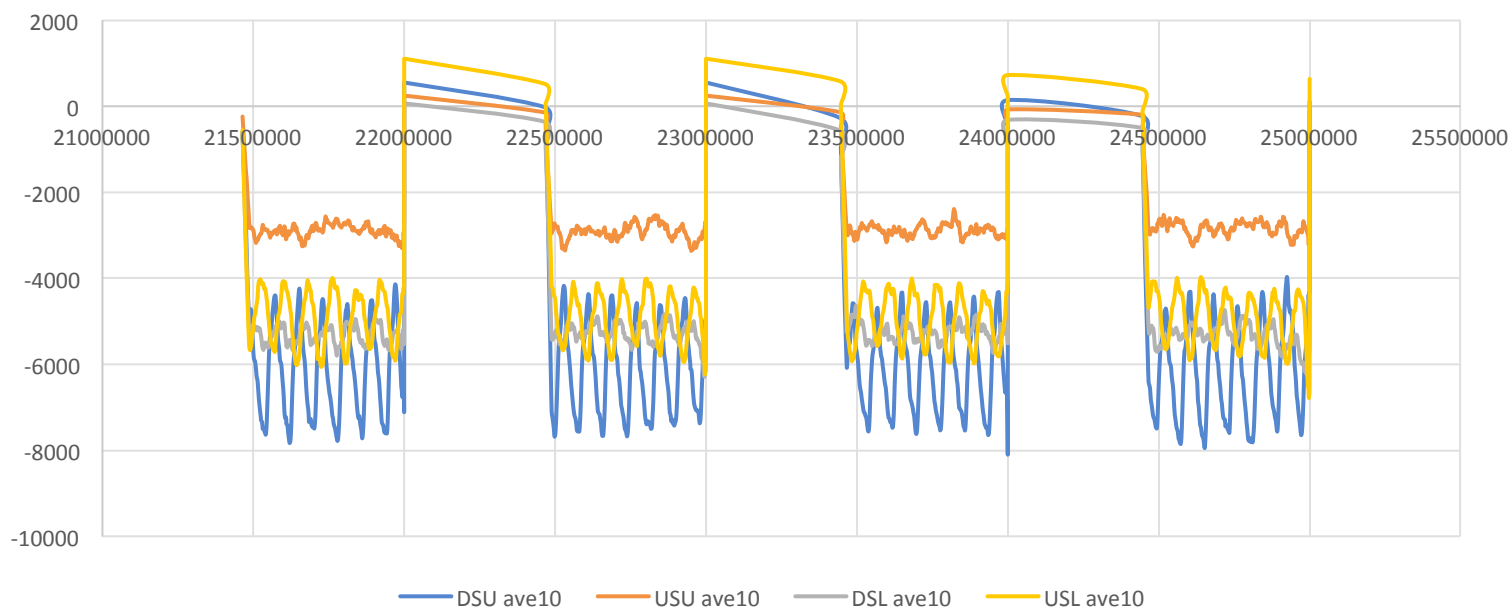
16-ID 3m IVU linear encoder data

Moving gap position from 24mm to 23 mm, 15 servo cycle gather period



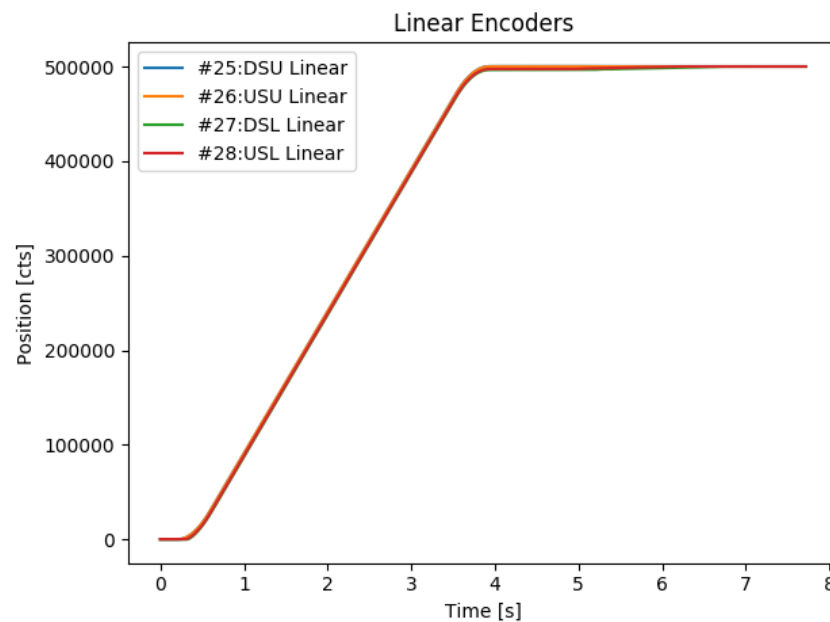
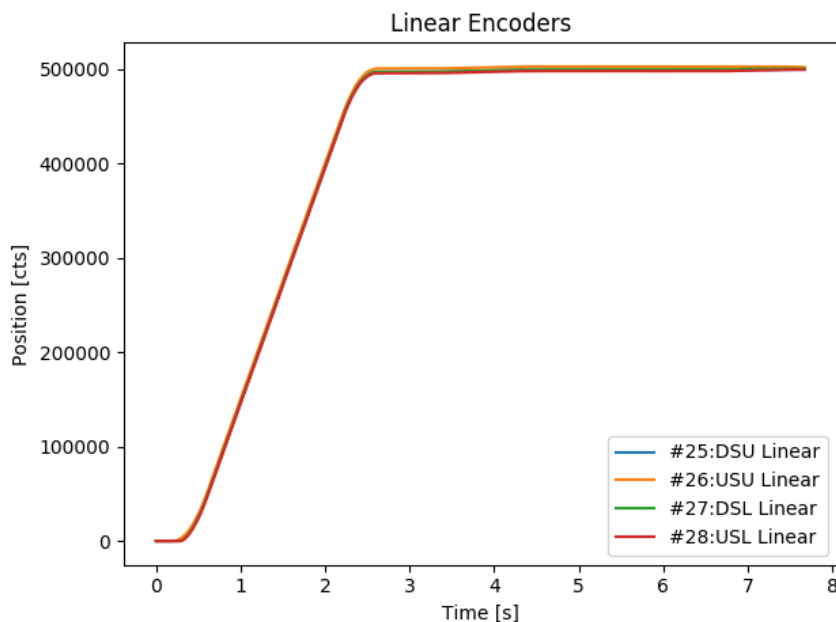


LIX torque from 25 to 21+ mm motor bolted back down centered horizontally.



Repeatability of IVU motion

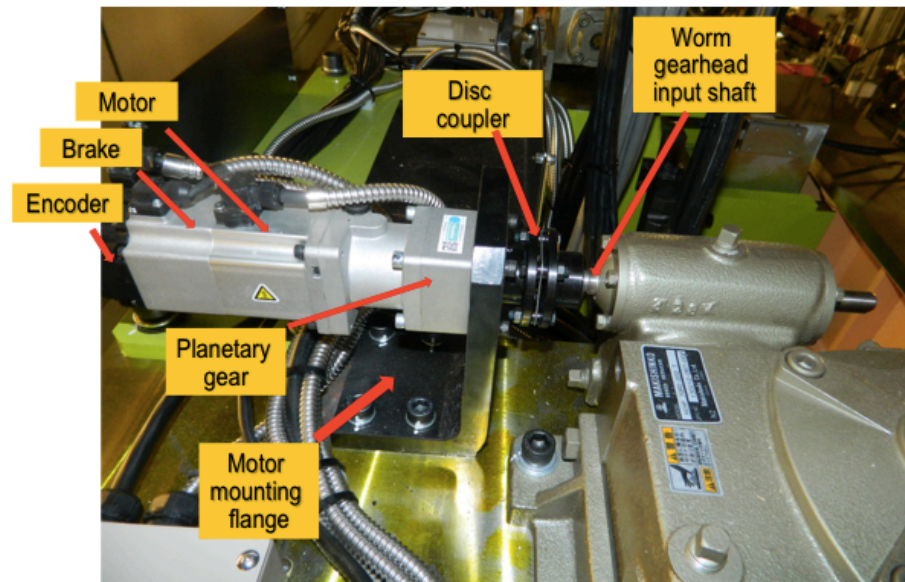
Starting gap at 25mm ending at 24mm



Currently under development

- Redesign of TPMAC source code for Neomax IVU at cells 4,12, and 16
- Mechanical repairs on undulators drive train
- EPU Phasing

1.5 meter IVU Drive train coupling



Acknowledgements

Special thanks to Insertion Device Task
Force at NSLS-II and PCaPAC 2018
Committee, Sponsors, and Organizers

Any Questions?